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Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

ANAESTHESIA IN PRE-ECLAMPSIA

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1. Introduction and Purpose

Pre-eclampsia is a complex multisystem disorder occurring after the 20th week of gestation involving hypertension, proteinuria and in severe cases cerebral symptoms or low platelets. Intravenous fluid should be administered very carefully since pulmonary oedema is a common iatrogenic complication.

2. Scope

This guideline is intended as a reference for trainee anaesthetists seeking advice regarding the management of anaesthesia in pre-eclampsia.

3. Points to consider

3.1 Regional anaesthesia in pre-eclampsia / HELLP

- Check the platelet count before performing an epidural / spinal block in all pre eclampsia patients (In severe pre eclampsia / HELLP, or a previous known unstable platelet count, the blood sample on which the platelet count is performed should have been taken no more than 2 hours prior to the spinal / epidural)
- If platelet count < 100 per nanolitre then perform a clotting screen
- If the platelet count is > 80 per nanolitre and stable, and the clotting screen is normal, it is then safe to perform an epidural
- If the platelet count is < 80 per nanolitre or clotting screen is abnormal then do not perform a regional technique

3.2 Anaesthesia for Caesarean Section in pre-eclampsia / HELLP

The presence of a senior anaesthetist is mandatory for cases of severe pre eclampsia / HELLP

3.2.1 Regional anaesthesia

- Spinal or epidural anaesthesia is the preferred mode of anaesthesia if the platelet count and clotting screen are normal
- Spinal (and epidural) anaesthesia has been shown to produce less hypotension in pre eclamptics than in normal parturients despite the former having a depleted intravascular volume
- Preload volume should be reduced to a maximum of 500ml, with colloid possibly being used in preference to crystalloid

- Epidural catheters should not be removed unless the platelet count is > 80 per nanolitre, and the clotting screen is normal.

3.2.2 General Anaesthesia

- Consider general anaesthesia if :

  i. No time for regional anaesthesia

  ii. Regional anaesthesia contraindicated e.g. low platelets

- Assess the airway carefully, and beware of facial oedema or stridor. (Consider awake fibreoptic intubation in severe cases).

- In very severe pre eclampsia invasive arterial monitoring should be established prior to induction.

- Gross hypertension should be controlled with 10 – 20 mg boluses of labetolol prior to induction.

- A modified rapid sequence technique should be used. The modification being the administration of alfentanil 15mcg /Kg (1mg usually used) following preoxygenation and immediately prior to induction. This attenuates the pressor response to intubation.

- If MgSO4 has been used, then a prolonged neuromuscular block can occur. Therefore reduce the dose of non depolarising muscle relaxant used and use a nerve stimulator.

- An indwelling urinary catheter with an urometer attached is essential for measuring hourly urine output in the perioperative period.

- Consider CVC line insertion for optimal fluid management. Avoid the subclavian route in coagulopathic patients.

- Ensure adequate analgesia prior to extubation

- Avoid NSAIDs as these patients are prone to impaired renal function and platelet count or function. NSAIDS can be commenced once the platelet count is >100 per nanolitre and proteinuria has resolved

- Postoperatively the patient should be transferred transfer to ICU if the patient needs to remain intubated and ventilated. Otherwise transfer to Obstetric HDU for further care and monitoring of vital parameters (Pulse, BP, CVP, temperature, oxygen saturation, respiratory rate, urine output and conscious level)

4. References

1. Brodie H., Manilow A. M. Anaesthetic management of pre-eclampsia /eclampsia. *International Journal of Obstetric Anesthesia* April 1999; 8:110 -124

2. Anaesthesia and pre-eclampsia / eclampsia. Norris M.C. (2000) *Handbook of Obstetric Anaesthesia. First Edition*.Lippincott Williams & Wilkins : 25 – 103